EXPERIMENT – 8

Privilege Escalation Using TryHackMe Platform

Aim

To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory (Brief)

Privilege Escalation is a technique used to gain higher-level permissions than originally assigned. In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.
TryHackMe provides a legal and controlled environment to practice privilege escalation techniques and understand real-world attack scenarios.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room

1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration

1. Check current user:

2. whoami
3. Check system information:
4. uname -a
5. Check sudo permissions:
6. sudo -l

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command:

find / -perm -4000 2>/dev/null

Step 6: Exploit Vulnerability

1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
4. whoami

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output

- Successful privilege escalation from normal user to root.
- System commands executed with elevated privileges.
- Screenshots captured as proof of completion.

```
 * Documentation:  https://help.ubuntu.com/

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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Last login: Fri Jun 18 04:38:27 2021 from 10.0.2.15
Could not chdir to home directory /home/karen: No such file or direct
ory
$ whoami
karen
$ uname -a
Linux wade7363 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 U
TC 2014 x86_64 x86_64 x86_64 GNU/Linux
$ sudo -l
[sudo] password for karen:
Sorry, user karen may not run sudo on wade7363.
$ sudo -l
[sudo] password for karen:
$ sudo -l
[sudo] password for karen:
Sorry, user karen may not run sudo on wade7363.
$
LineKernel                                                    57min 8s
```

```
$ find / -perm -4000 2>/dev/null
/usr/sbin/pppd
/usr/sbin/uuidd
/usr/lib/eject/dmcrypt-get-device
/usr/lib/policykit-1/polkit-agent-helper-1
/usr/lib/pt_chown
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/lib/x86_64-linux-gnu/oxide-qt/chrome-sandbox
/usr/bin/chsh
/usr/bin/chfn
/usr/bin/mtr
/usr/bin/lppasswd
/usr/bin/sudo
/usr/bin/X
/usr/bin/passwd
/usr/bin/pkexec
/usr/bin/newgrp
/usr/bin/gpasswd
/usr/bin/traceroute6.iputils
```

```
/proc/950/attr/exec
/proc/950/attr/fscreate
/proc/950/attr/keycreate
/proc/950/attr/sockcreate
/proc/975/task/975/attr/current
/proc/975/task/975/attr/exec
/proc/975/task/975/attr/fscreate
/proc/975/task/975/attr/keycreate
/proc/975/task/975/attr/sockcreate
/proc/975/attr/current
/proc/975/attr/exec
/proc/975/attr/fscreate
/proc/975/attr/keycreate
/proc/975/attr/sockcreate
/proc/992/task/992/attr/current
/proc/992/task/992/attr/exec
/proc/992/task/992/attr/fscreate
/proc/992/task/992/attr/keycreate
/proc/992/task/992/attr/sockcreate
/proc/992/task/1015/attr/current
/proc/992/task/1015/attr/exec
/proc/992/task/1015/attr/fscreate
/proc/992/task/1015/attr/keycreate
/proc/992/task/1015/attr/sockcreate
/proc/992/task/1101/attr/current
/proc/992/task/1101/attr/exec
/proc/992/task/1101/attr/fscreate
/proc/992/task/1101/attr/keycreate
/proc/992/task/1101/attr/sockcreate
/proc/992/task/1102/attr/current
/proc/992/task/1102/attr/exec
/proc/992/task/1102/attr/fscreate
/proc/992/task/1102/attr/keycreate
/proc/992/task/1102/attr/sockcreate
/proc/992/attr/current
LineKernel                                                    55min 23s
```

```
$ cat /etc/crontab
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab'
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin

# m h dom mon dow user  command
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || ( cd / && run-pa
rts --report /etc/cron.daily )
47 6    * * 7   root    test -x /usr/sbin/anacron || ( cd / && run-pa
rts --report /etc/cron.weekly )
52 6    1 * *   root    test -x /usr/sbin/anacron || ( cd / && run-pa
rts --report /etc/cron.monthly )
#
$
```

Result

Thus, privilege escalation was successfully performed using the TryHackMe platform.
The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.

⚠️ Important Note

This experiment is performed **strictly for educational and ethical purposes** using the TryHackMe learning platform.